

KACPER KORBAN

PhD Student / Software Engineer

@ kacper.korban@epfl.ch

+41 783 378 783

korban.dev

kacperfkorban

KacperFKorban

EDUCATION

PhD in Computer Science

2024 - Ongoing

École Polytechnique Fédérale de Lausanne (EPFL)

Research area: Programming Languages and Formal Methods

Advisor: Clément Pit-Claudel, Systems and Formalisms Laboratory

M.Sc. in Computer Science

2021 - 2023

Chalmers University of Technology

GPA: 4.9/5.0

Thesis: "Verified Function Inlining Optimization for the PureCake Compiler"

Supervised by: Magnus O. Myreen

B.Sc. in Computer Science

2017 - 2021

AGH University of Science and Technology

Final Grade: 5.0/5.0, GPA: 4.69/5.0

Thesis: "A plugin for Dokka generating a database that allows for performing semantic queries"

Supervised by Bartosz Baliś

PUBLICATIONS



Lorikeet: Flexible code rewriting for Scala 3

Scala Workshop 2026

Sidonie Bouthors, Kacper F. Korban, Clément Pit-Claudel



Verified Inlining and Specialisation for PureCake

ESOP 2024

Hrutvik Kanabar, Kacper F. Korban, Magnus O. Myreen

EXPERIENCE

Software Developer

July 2019 - Aug 2024

VirtusLab

Federated SQL Compiler

Nov 2023 - Aug 2024

Developing a federated SQL compiler for executing queries between different big data warehouses for AdTech domain.

Scala SQL Spark BigData ZIO Kotlin JVM

Besom (Pulumi Scala)

July 2023 - Nov 2023

Working on Pulumi implementation for Scala (Besom) and its integration with Scala effect libraries.

Scala 3 Cloud Infrastructure as Code Functional Programming

Scala 3 Compiler and Open Source

Feb 2021 - July 2023

Developing and maintaining the Scala 3 compiler, Scaladoc, magnolia and other open source libraries.

Scala 3 Compilers Programming Languages Code Generation

Manufacturing Management System

July 2019 - Feb 2021

Full-stack development and maintenance of a management system for manufacturing companies.

Scala REST PostgreSQL Cats TypeScript Angular

STUDENT PROJECT SUPERVISION



Autoformalization of programming-language standard updates

2026

Valentin Schneeberger

AI Rocq



Augmenting EasyCrypt with an LSP Server and Alectryon support

2026

Roman Batut & Tristan Cendan

LSP Alectryon Easycrypt

 **Lorikeet: Flexible code rewriting for Scala**  2025 – 2026
Sidonie Bouthor

Scala Program rewriting Software migration Student feedback

 **Early reclamation for lazy values in Scala 3**  2025 – 2026
Guillaume Carraux

Scala 3 Compilers JVM Static analysis

OPEN SOURCE / SIDE PROJECTS

Metals – Scala language server and client

- Contributed several new features, including end-marker inlay hints and new code actions

Scala Developer tooling

VsRocq – Rocq language server and client

- Contributed bug fixes and new features, most notably an MCP server implementation

Rocq Developer tooling AI

Inkuire – Type-based search engine for Scala 3

Type based function search engine for Scala and Kotlin.

Programming languages Search Algorithms Scala Kotlin

AvocADO – Safe compile-time parallelization of for-comprehensions

Library providing compile-time rewrite for execution parallelization.

Scala 3 Functional Programming Compile-time reflection

QuickLens – Minimalistic lens library for Scala

- Developed the new implementation of the library for Scala 3 using Quotes metaprogramming API.

Scala 3 Lenses Functional Programming Compile-time reflection

TALKS

 **POPL'26 Student Research Competition**  2026
"Bootstrapping a verified compiler for an imperative language in Rocq"

 **Scala Days**  2025
"On track with tracked"

 **Wrocław Scala User Group**  2024
"Scala 3 features you probably haven't used (yet)"

 **Scala Love**  2022
"Scala 3, but I have trust issues"

 **Kotlin micro.sphere.it**  2021
"Inkuire - Using types as search keys in Kotlin libraries"

ACHIEVEMENTS

 **2nd place at POPL'26 SRC**  2026
"Bootstrapping a verified compiler for an imperative language in Rocq"

 **AGH Scholarship**  2018, 2020, 2021
Scholarship awarded to 10% of best students at AGH UST

 **SIP-62 - better fors**
Implementation of several improvements to for-comprehensions in Scala 3

TEACHING / KNOWLEDGE SHARING

	Teaching Assistant for CS-214 (EPFL) Mostly responsible for organizing the final project where students implemented web applications in groups. This also included deploying the applications in a global instance.	 2024, 2025
	Algorithms Teacher Teaching younger students Introduction to Algorithms at BIT Scientific Group	 2018 – 2020
	Company Tech Blog Writing posts for the company's tech blog	 2021 – 2023

TECH INTERESTS

- Formal Verification
- Compilers
- Interactive Theorem Proving
- Programming Languages
- Developer Tooling

LANGUAGES

Polish	
English	
German	